

ΕΘΝΙΚΟ ΜΕΤΣΟΒΙΟ ΠΟΛΥΤΕΧΝΕΙΟ
ΤΜΗΜΑ ΕΦΑΡΜΟΣΜΕΝΩΝ ΜΑΘΗΜΑΤΙΚΩΝ ΚΑΙ
ΦΥΣΙΚΩΝ ΕΠΙΣΤΗΜΩΝ, ΤΟΜΕΑΣ ΜΗΧΑΝΙΚΗΣ
ΠΟΛΥΤΕΧΝΕΙΟΥΠΟΛΗ ΖΩΓΡΑΦΟΥ, ΚΤΙΡΙΟ ΘΕΟΧΑΡΗ
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**Re.: Dr Fernando Alonso-Marroquin, APPLICANT FOR VACANCY
NUMBER: 152633. Lecturer/Senior Lecturer in Geomechanics/Geotechnical
Engineering position at The University of Sydney.**

Date: Sunday, June 14, 2009

Dear Ms Paulina Rojas

With this letter I would like to report to you on the suitability of Dr. Fernando Alonso-Marroquin for the Lecturer position in Geotechnical Engineering.

I met Dr Alonso-Marroquin as he worked in my laboratory as a Young post doctoral Researcher employed by the European RTN DIGA (EU Research & Training Network on “Degradation and Instabilities of Geomaterials Applied to hazard mitigation”) that was coordinated by myself. Within this project he has performed excellently and he was very productive in terms of scientific results reported in high quality research publications. **During his stay at Athens he had been in close contact with my PhD students and he supervised in particular the Master thesis of one of my students, Mr. Ioannis Stefanou, on the numerical modeling of granular flow.**

Dr Alonso-Marroquin **has 29 publications in the area of computational geomechanics (14 published paper in Journals and 15 in pre-refereed Conference Papers).** According to Google Scholar his publications have **more than 100 citations.** This is to my judgment a very good standing for an Early Carrier Researcher.

Dr Marroquin is already internationally known through his invited participation in International Symposia and specialty in workshops on granular material modeling and through his contributions to the discrete element modeling of granular media and the

connection between micromechanics and continuum mechanics. He has a well established research area in the development of novel techniques to represent realistic particle shape and contact interactions in 3D discrete element simulations. More recently he has established a research group of his own, which focuses in the numerical modeling of important problems of geomechanics and geosciences (investigation of the weakness of fault zones, the improvement of efficiency of granular flow, development of climate models for ice melting and hurricanes formation, and coupling between different numerical modes for multiphysics simulations)

Dr. Alonso-Marroquin is highly motivated, a team worker, a dedicated teacher, and above a critical and independent mind. The position as Lecturer in Geotechnical Engineering will be very positive for the development of a broader program in research and teaching in the multi-scale modeling of geomaterials. This will allow him to establish new research projects with experimentalist and practitioners of the School of Civil Engineering in key areas **of fundamental and applied geotechnical engineering.**

In summary I would like to recommend enthusiastically Dr. Fernando-Alonso Marroquin for the applied position in your department.

Sincerely

Ioannis Vardoulakis

Professor of Mechanics